

Walsh-Hadamard Transform and p-Adic (Dis)continuity in Collatz Dynamics

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Abstract: We introduce the Walsh-Hadamard Transform (WHT) into Collatz research, establishing it as the character group of 2-adic integers modulo 2^K . We derive exact closed forms for the ramp signal, the folding formula, and the piecewise-constant support theorem. Concurrently, we refute the 3-adic continuity of σ via hierarchical mean tests and quantify a striking cross-character asymmetry: the 3-adic lattice energy is three orders of magnitude lower than its 2-adic counterpart, providing rigorous criteria for selecting appropriate harmonic analysis tools.

1. WHT Closed Forms and 2-Adic Characters (S-I1~I4)

Theorem 5 (WHT Closed Forms) For the ramp signal $\sigma(2^k)=k$ ($N=2^K$), the WHT support is exactly $K+1$ lines:

$$\begin{aligned} W[q] &= N(N+1)/2 \quad (q=0), \\ W[2^b] &= -N \cdot 2^{b-1} \quad (1 \leq b \leq K-1), \text{ and } 0 \text{ otherwise.} \end{aligned}$$

Folding formula: $W(\sigma)[q] = W(e)[q \gg 1] \pm W(o)[q \gg 1]$, exactly separating the odd half (3-expansion domain) from the even half (dyadic domain). The odd half carries 55.67% of the total energy in the Walsh basis — the Walsh-domain quantification of the 3-expansion difficulty.

2. Refutation of 3-Adic Continuity (S-J2)

If σ were 3-adically continuous (class means converging mod 3^j), it would imply spectral regularity. Our tests reveal:

- The range of class means mod 3^j grows with j (slope $+0.886$, $\gamma = -0.886 < 0$);
- The argmax class drifts as $[2, 0, 9, 9, 20]$ for mod $3/9/27/81/243$, without a fixed 3-adic limit;

- The range at 3^5 (9.16) is $\approx 3.5\times$ the sampling noise.

Conclusion: Ramanujan-type 3-adic expandability fails; 3-adic harmonic analysis is unsuitable for decomposing σ .

3. Cross-Basis Character Spectrum Asymmetry (S-L5)

On the group $\mathbb{Z}/3^9$ ($N=19683$), the 3-adic lattice (bins that are multiples of $N/3$) contains only $3.0e-5$ of the total energy — comparable to Gaussian noise ($3.9e-5$). In contrast, the analogous 2-adic lattice on a 2-adic window holds 0.033, a **three-order-of-magnitude difference**. This quantitatively proves that σ 's spectral structure is purely 2-adic; the 3-adic character basis reveals no point structure.

Keywords: Walsh-Hadamard · 2-adic character · 3-adic discontinuity · cross-basis asymmetry · S-I/S-J